

Ardie Barry Sails

PhD Candidate in Pharmaceutical Sciences (Health)

Department of Pharmaceutical Life Sciences, Faculty of Pharmacy, Universiti Malaya

ardiebarrys@gmail.com ardiebarrysails.com [ORCID 0009-0009-8994-2793](https://orcid.org/0009-0009-8994-2793) [Google Scholar](https://scholar.google.com/) [linkedin.com/in/ardiebarrysails](https://www.linkedin.com/in/ardiebarrysails)

Kuala Lumpur, Malaysia

Profile

Toxicology researcher studying how e-cigarette use affects male reproductive health. My PhD at Universiti Malaya compares cigarette smokers, e-cigarette users and dual users in Malaysia, linking exposure to testosterone-related microRNAs, mitochondrial function and blood cell changes. I have published 13 peer-reviewed papers as sole or first author and review manuscripts for seven international journals. Open to roles in research, medical affairs, regulatory science and science communication.

Education

Doctor of Philosophy in Pharmacy (Health)

Dec 2023 – Dec 2026 (expected)

Universiti Malaya, Kuala Lumpur

Thesis on testosterone-related microRNAs, blood cell abnormalities and sexual desire in Malaysian cigarette smokers, e-cigarette users and dual users.

Bachelor of Science in Microbiology, With Distinction

2019 – 2023

Universiti Malaya, Kuala Lumpur

Experience

Doctoral Researcher

Dec 2023 – present

Department of Pharmaceutical Life Sciences, Faculty of Pharmacy, Universiti Malaya

- Conducted a comparative study of cigarette smokers, e-cigarette users and dual users in Malaysia.
- Analyzed testosterone-related microRNAs by qPCR alongside hormone and blood cell profiles.
- Used transmission electron microscopy to assess mitochondrial damage in e-cigarette users and controls.
- Wrote mechanistic and systematic reviews on e-cigarette toxicology, reproductive health and redox signaling.
- Started an independent project, Cellular Signalling as Dynamic Regulatory Circuits, which produced the redoxostat concept in NRF2–KEAP1 biology.

Peer Reviewer

2025 – present

Seven international journals

Reviews manuscripts in toxicology, pharmacology, reproductive health and public health for American Journal of Preventive Medicine, Journal of Hazardous Materials Advances, Pharmacological Reviews, Progress in Biophysics and Molecular Biology, Toxicology Reports, PLOS One and International Journal of General Medicine.

Supported laboratory operations during a full-time internship, covering microbiology procedures, water quality testing, sample handling, documentation and technical reporting under standard operating procedures.

Publications

13. Transcriptional condensates as kinetic filters for temporal control of gene expression.

Ardie Barry Sailis

Gene Reports, 2026. doi.org/10.1016/j.genrep.2026.102599

12. Mitochondrial mechanisms in the potential male reproductive toxicity of e-cigarette-derived metal-containing nanoparticles.

Ardie Barry Sailis

Journal of Applied Toxicology, 2026. doi.org/10.1002/jat.70382

11. Ferroptosis in e-cigarette aerosol-associated respiratory injury.

Ardie Barry Sailis

Archives of Toxicology, 2026. doi.org/10.1007/s00204-026-04486-w

10. Could e-cigarette devices generate inhalable micro- and nanoplastics? Exposure plausibility and reproductive relevance.

Ardie Barry Sailis

Toxicology Mechanisms and Methods, 2026. doi.org/10.1080/15376516.2026.2695155

9. Adipose as a Driver, Not a Bystander: A Modern Synthesis of Obesity-Related Erectile Dysfunction.

Ardie Barry Sailis, Muhamad Alfakri Mat Noh

Diabetes, Obesity and Metabolism, 2026. doi.org/10.1111/dom.70818

8. E-cigarette aerosol constituents modulate Leydig cell steroidogenic pathways: Evidence from experimental models.

Ardie Barry Sailis, Muhamad Alfakri Mat Noh, Bey Fen Leo, Farid Nazer Faruqu, Anne Yee, Maw Shin Sim

Molecular and Cellular Endocrinology, 2026. doi.org/10.1016/j.mce.2026.112786

7. MicroRNA-mediated disruption of testosterone signaling associated with e-cigarette exposure.

Ardie Barry Sailis, Muhamad Alfakri Mat Noh, Bey Fen Leo, Farid Nazer Faruqu, Anne Yee, Maw Shin Sim

Environmental Toxicology and Pharmacology, 2026. doi.org/10.1016/j.etap.2026.104994

6. NRF2-KEAP1 as a redox signal-resolution circuit: Beyond the antioxidant switch.

Ardie Barry Sailis

Progress in Biophysics and Molecular Biology, 2026. doi.org/10.1016/j.pbiomolbio.2026.03.005

5. E-cigarettes and erectile dysfunction: biological mechanisms and research challenges.

Ardie Barry Sailis, Muhamad Alfakri Mat Noh, Bey Fen Leo, Farid Nazer Faruqu, Hui Yin Yow, Anne Yee, Maw Shin Sim

International Journal of Impotence Research, 2026. doi.org/10.1038/s41443-026-01300-0

4. E-cigarette aerosols as systemic metabolic disruptors: integrated mitochondrial, circadian, and neurobehavioral mechanisms.

Ardie Barry Sailis

Toxicology Mechanisms and Methods, 2026. doi.org/10.1080/15376516.2026.2658739

3. CYP1A1 as a conserved metabolic circuit linking environmental sensing to immune regulation.
Ardie Barry Sailis
Archives of Toxicology, 2026. doi.org/10.1007/s00204-026-04384-1

2. Effects of secondhand exposure to e-cigarette aerosol on lung health: a systematic review.
Ardie Barry Sailis, Muhamad Alfakri Mat Noh, Bey Fen Leo, Farid Nazer Faruqu, Anne Yee, Maw Shin Sim
Journal of Public Health, 2026. doi.org/10.1007/s10389-026-02740-0

1. Mitochondrial dysfunction induced by E-cigarettes.
Ardie Barry Sailis, Muhamad Alfakri Mat Noh, Bey Fen Leo, Farid Nazer Faruqu, Anne Yee, Maw Shin Sim
Toxicology, 2025. doi.org/10.1016/j.tox.2025.154339

Talks and conferences	PhD thesis seminar	1 Oct 2026
	Universiti Malaya	
	Sexual Desire and Nicotine Dependence Among Smokers, E-Cigarette Users, and Dual Users in Malaysia: A Comparative Study	
	2024 Poster presentation, ISAM Regional Meeting	
	Early Career Researchers–Postgraduates Research Symposium	2024
	ic-PHIR Conference	2023
	UiTM Puncak Alam	

Training	Systematic Literature Review and Meta-Analysis Workshop	2025
	UV-Visible Double Beam Spectrophotometer Workshop, Shimadzu	2025
	Quantitative PCR (qPCR) Workshop, Vazyme, Universiti Malaya	2024
	ic-PHIR Metabolic Workshop, UiTM Puncak Alam	2023

Skills	Laboratory	qPCR, transmission electron microscopy, molecular biology techniques
	Data	SPSS, bioinformatics, data interpretation
	Evidence	Systematic reviews, literature synthesis, mechanistic toxicology
	Writing	Scientific writing, manuscript preparation, peer review, research conceptualization
	Research areas	E-cigarette and tobacco toxicology, reproductive health, pharmacology, toxicogenomics, molecular systems biology

Languages	Malay	Native or bilingual proficiency
	English	Professional working proficiency

References	Available on request.
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